

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (E & C) Examination

Nov/Dec 2013

EC – 406 Digital Image Processing

Date: 23.11.13, Saturday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question number.
- (3) Figures to right indicate marks
- (4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

- Q-1 A Explain in detail the connectivity of pixels. Consider the image with subsets S_1 [6]
and S_2 shown in figure below. For $V = \{2\}$, determine whether S_1 and S_2 are
(a) 4-connected, (b) 8-connected (c) m-connected.
Also calculate the city block distance, chess board distance and m-adjacency
distance between the pixels encircled in the image.

S_1				S_2			
0	0	0	0	0	0	0	2
0	0	2	0	0	②	0	0
0	0	2	0	2	2	0	0
0	②	2	2	0	0	0	0

- B Answer following questions. [5]
- (i) What is dynamic range?
 - (ii) Differentiate photopic and scotopic vision.
 - (iii) Find the number of bits required to store a 256 X 256 image with 32 gray levels.
 - (iv) What is resolution? Explain the types of resolution.
 - (v) Define Weber ration.

- Q-2 Attempt any three.

- A Explain the method to calculate the gradient of color image. [4]
B The Laplacian of an image $f(x,y)$ is defined as

$$\nabla^2 f(x,y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \quad [4]$$

Prove that fourier transform of this Laplacian is given by

$$F\{\nabla^2 f(x,y)\} \Leftrightarrow (-2\pi)^2(u^2 + v^2)F(u,v).$$

- C Describe the homomorphic filtering and its advantage. [4]
D Explain the various low pass frequency domain filter and explain the role of radius in term of image total power. [4]

Q-3 Attempt any three.

- A Given $h(x,y)$, discuss the frequency response

[4]

-1	-1	-1
-1	8	-1
-1	-1	-1

- B Describe the HSI model in detail. For RGB Value (100,150,200), Compute equivalent HSI value. [4]

- C What are the various component of image processing? [4]

- D Determine the fourier transform of following image: [4]

0	1	2	1
1	2	3	2
2	3	4	3
1	2	3	2

SECTION – II

- Q-4 A Explain the image formation model and various image processing applications. [6]
 B Given below a 3 X 3 image, what will be the value of the centre pixel change to when this image is passed through [5]

- (i) Arithmetic mean filter
 (ii) Geometric mean filter
 (iii) Harmonic filter
 (iv) Max filter
 (v) Min filter

1	7	5
6	2	3
1	4	2

- Q-5 Attempt any two.

- A Equalize the given histogram and plot it. What happens if equalized the image histogram twice? [6]

Gray level	0	1	2	3	4	5	6	7
Number of pixels	790	1023	850	656	329	245	122	81

- B With reference to contraharmonics filter [6]
 (i) Explain how filter is effective in eliminating salt and pepper noise.
 (ii) Explain why filter gives poor results if wrong polarity is selected for Q.
 (iii) Discuss the behavior of filter when $Q = -1$.
 (iv) Discuss (for positive and negative Q) the behavior of filter in areas of constant gray levels.
 C What is Piecewise linear transformation functions? Explain any one function in brief. [6]

- Q-6 Attempt any three.

- A Considering the image of 4 bit with size of 4 X 4 shown below, Explain the effect on the histogram, if (i) higher two bitplanes set to zero and (ii) lower two bitplanes set to zero. [4]

0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

- B Derive mask (size of 3×3) of second derivatives by Laplacian, for enhancement in spatial domain. Prepare the other modified mask of the same. [4]
 C What is blind deconvolution in estimation of degradation model? What are the principles of estimation? [4]
 D Compare noise reduction adaptive filters with order statistic filters. [4]